



US012418484B2

(12) **United States Patent**
Mohanty et al.

(10) **Patent No.:** **US 12,418,484 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **SYSTEMS AND METHODS FOR
DETERMINING NETWORK COMPONENT
SCORES USING BANDWIDTH CAPACITY**

(58) **Field of Classification Search**

None

See application file for complete search history.

(71) Applicant: **Cisco Technology, Inc.**, San Jose, CA
(US)

(56) **References Cited**

U.S. PATENT DOCUMENTS

(72) Inventors: **Satya Ranjan Mohanty**, San Ramon,
CA (US); **Mankamana Prasad
Mishra**, Dublin, CA (US); **Ali Sajassi**,
San Ramon, CA (US); **Mrinmoy
Ghosh**, Milpitas, CA (US)

8,650,286 B1 * 2/2014 Sajassi H04L 45/18
709/224

9,571,570 B1 2/2017 Mutnuru
(Continued)

(73) Assignee: **CISCO TECHNOLOGY, INC.**, San
Jose, CA (US)

OTHER PUBLICATIONS

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

Sajassi, A. et al., "BGP MPLS-Based Ethernet VPN," Internet
Engineering Task Force (IETF), <https://tools.ietf.org/html/rfc7432>,
dated Feb. 2015, 56 pages.

(Continued)

(21) Appl. No.: **18/177,971**

Primary Examiner — Jenkey Van

(22) Filed: **Mar. 3, 2023**

(74) Attorney, Agent, or Firm — Baker Botts L.L.P.

(65) **Prior Publication Data**

US 2023/0208737 A1 Jun. 29, 2023

Related U.S. Application Data

(63) Continuation of application No. 17/857,861, filed on
Jul. 5, 2022, now abandoned, which is a continuation
(Continued)

(51) **Int. Cl.**
H04L 49/25 (2022.01)
H04L 12/18 (2006.01)

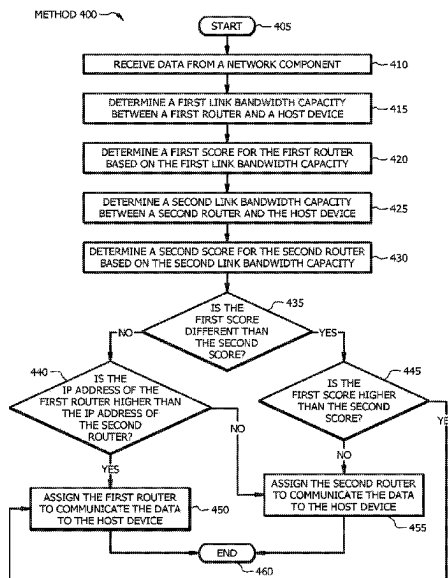
(Continued)

(52) **U.S. Cl.**
CPC **H04L 47/125** (2013.01); **H04L 12/18**
(2013.01); **H04L 12/4641** (2013.01);
(Continued)

(57) **ABSTRACT**

In one embodiment, a method includes receiving, by a first router, data from a network component. The method also includes determining, by the first router, a first link bandwidth capacity between the first router and a host device and determining, by the first router, a first score for the first router based on the first link bandwidth capacity. The method also includes determining, by the first router, a second link bandwidth capacity between a second router and the host device and determining, by the first router, a second score for the second router based on the second link bandwidth capacity. The method further includes comparing, by the first router, at least the first score and the second score to determine a highest score and assigning, by the first router, an edge router associated with the highest score to communicate the data to the host device.

30 Claims, 4 Drawing Sheets



Related U.S. Application Data

of application No. 16/696,203, filed on Nov. 26, 2019, now Pat. No. 11,394,632.

- (60) Provisional application No. 62/843,108, filed on May 3, 2019.

(51) **Int. Cl.**

H04L 12/46 (2006.01)

H04L 12/66 (2006.01)

H04L 41/0896 (2022.01)

H04L 43/0882 (2022.01)

H04L 45/00 (2022.01)

H04L 47/125 (2022.01)

H04L 69/326 (2022.01)

(52) **U.S. Cl.**

CPC **H04L 12/66** (2013.01); **H04L 41/0896** (2013.01); **H04L 43/0882** (2013.01); **H04L 45/22** (2013.01); **H04L 49/25** (2013.01); **H04L 69/326** (2013.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2015/0003450	A1 *	1/2015	Salam	H04L 12/1836 370/390
2015/0067033	A1	3/2015	Martinsen et al.	
2017/0013508	A1 *	1/2017	Pallas	H04W 28/088
2018/0287946	A1	10/2018	Nagarajan et al.	
2020/0280507	A1	9/2020	Chen et al.	
2021/0328936	A1 *	10/2021	Liu	H04L 45/16

OTHER PUBLICATIONS

Rabadan, J. et al., "Framework for Ethernet VPN Designated Forwarder Election Extensibility," <https://tools.ietf.org/html/rfc8584>, dated Apr. 2019, 32 pages.

Malhotra, N. et al., "Weighted Multi-Path Procedures for EVPN All-Active Multi-Homing draft-ietf-bess-evpn-unequal-lb-00," <https://tools.ietf.org/html/draft-ietf-bess-evpn-unequal-lb-00>, dated Sep. 19, 2018, 18 pages.

Mohanty, S. et al., "Weighted HRW and its applications draft-mohanty-bess-weighted-hrw-00," <https://tools.ietf.org/html/draft-mohanty-bess-weighted-hrw-00>, dated Mar. 11, 2019, 11 pages.

Mohapatra, P. et al., "BGP Link Bandwidth Extended Community draft-ietf-idr-link-bandwidth-07.txt," <https://tools.ietf.org/html/draft-ietf-idr-link-bandwidth-07>, Mar. 5, 2018, 5 pages.

Gredler, H. et al., "North-Bound Distribution of Link-State and TE Information using BGP draft-ietf-idr-ls-distribution-05," <https://tools.ietf.org/html/draft-ietf-idr-ls-distribution-05>, May 21, 2014, 36 pages.

Resch, James, "New Hashing Algorithms for Data Storage", Cleversafe, Storage Developer Conference, SNIA, Santa Clara, 2015, https://www.snia.org/sites/default/files/SDC15_presentations/dist_sys/Jason_Resch_New_Consistent_Hashings_Rev.pdf (Year: 2015).

Kamber, M., Pei, J., Han, J. (2011). Data Mining: Concepts and Techniques. Netherlands: Elsevier Science. (Year: 2011).

Stewart, James (2003). Calculus: early transcendentals. (Year: 2003). International Preliminary Report on Patentability for International Application No. PCT/US2020/030344, mailed Nov. 18, 2021, 9 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2020/030344, mailed Jun. 23, 2020, 10 Pages.

Office Action for European Application No. 207282799, dated Nov. 28, 2023, 5 Pages.

Rabadan J., Ed., et al., "Framework for EVPN Designated Forwarder Election Extensibility," draft-ietf-bess-evpn-df-election-framework-09, BESS Workgroup, Internet Draft, Updates: 7432, Intended status: Standards Track, Jan. 24, 2019, 29 Pages.

* cited by examiner

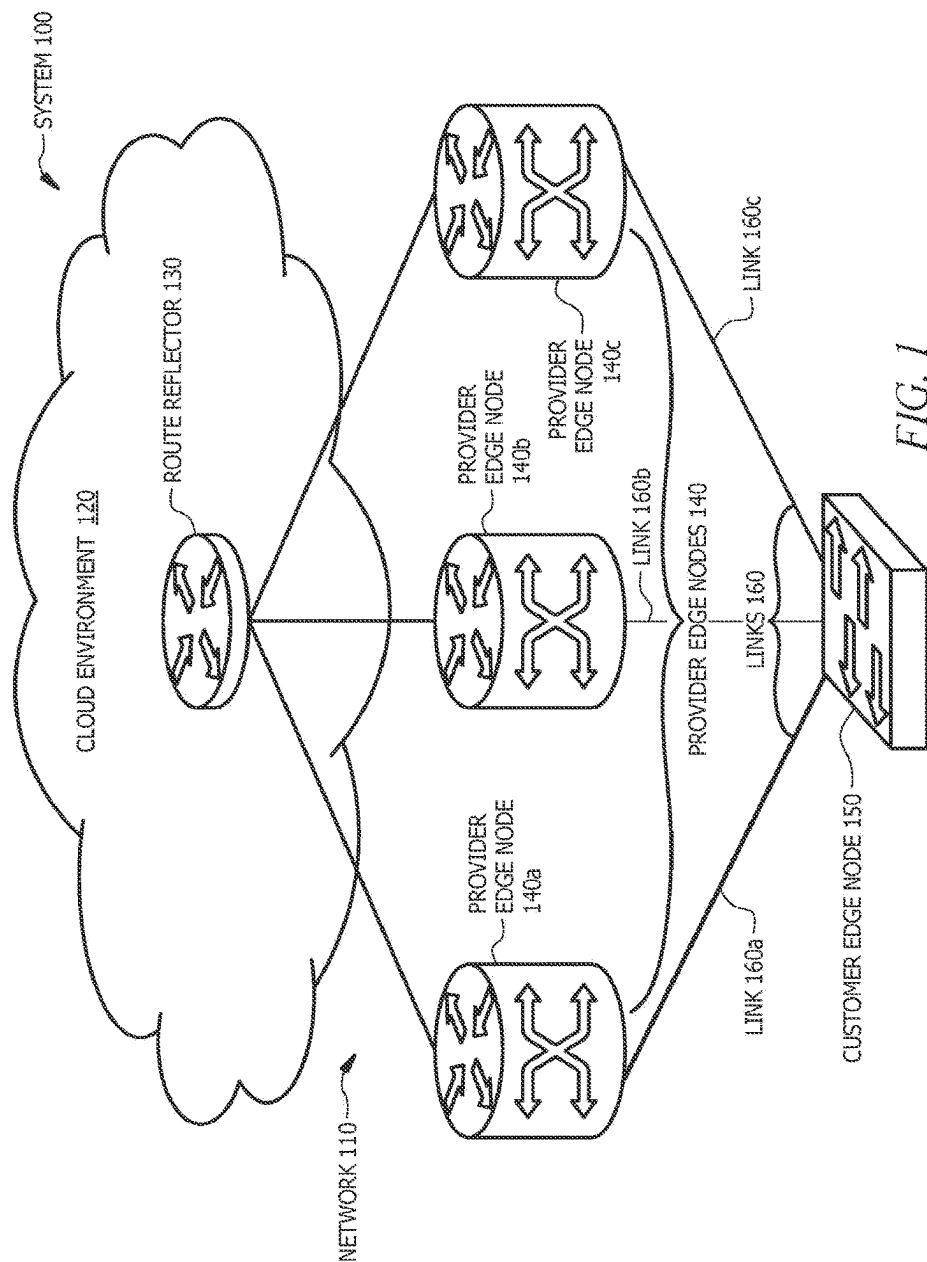


FIG. 1

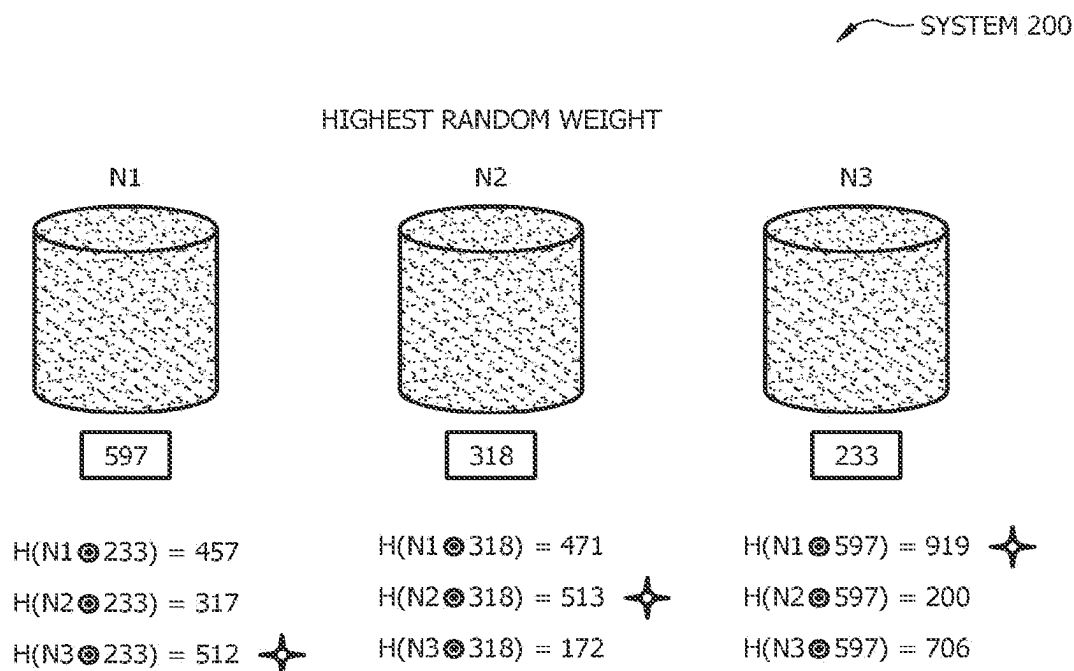


FIG. 2

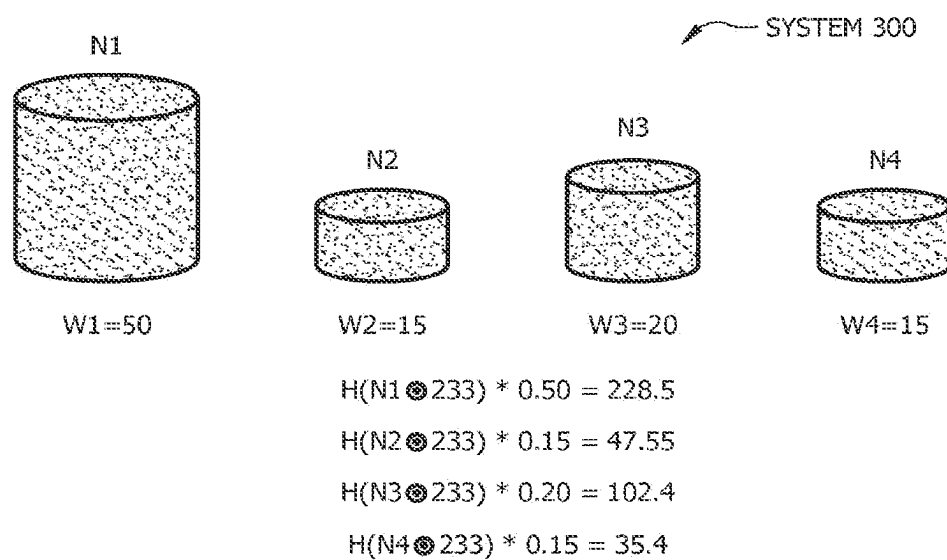


FIG. 3

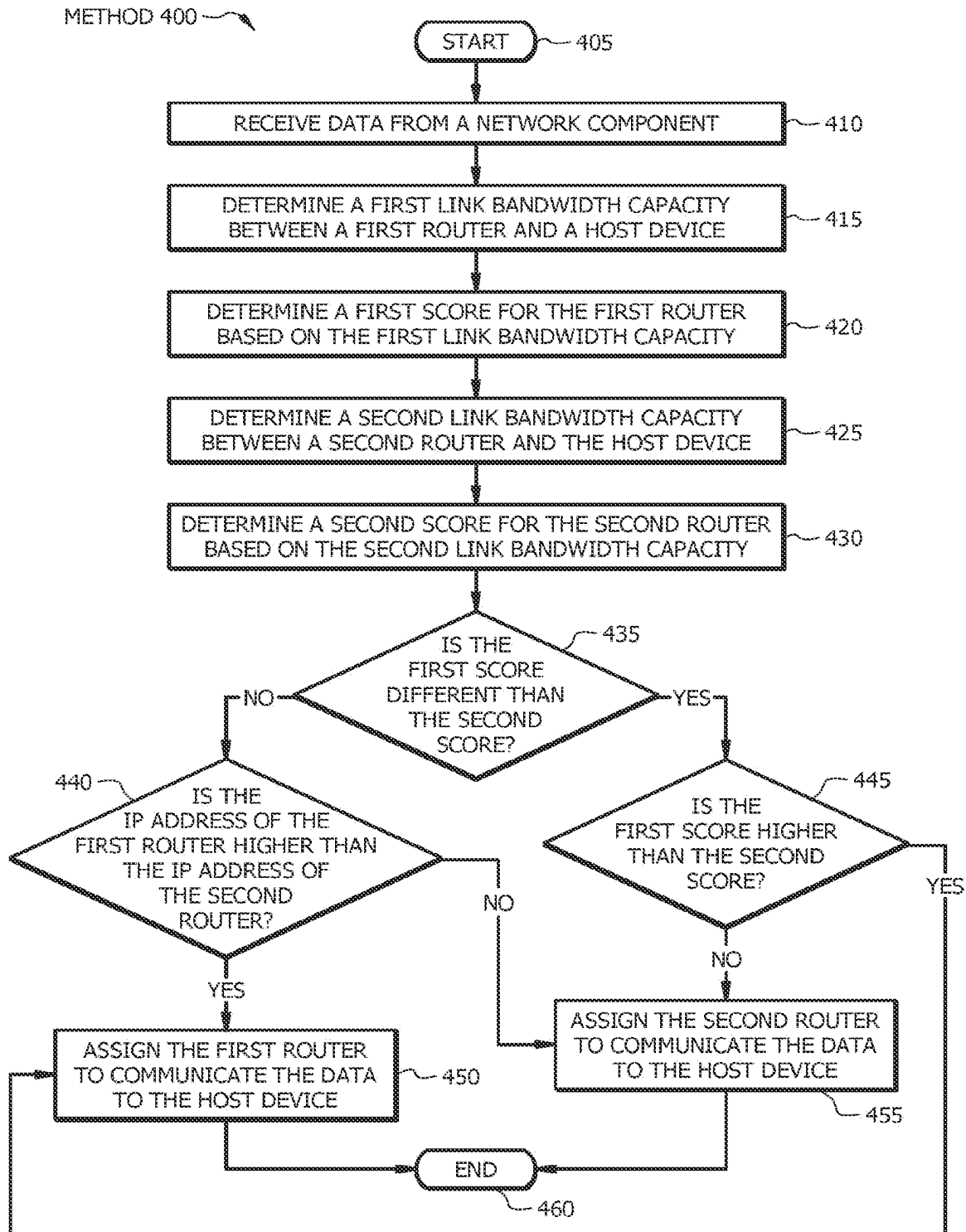
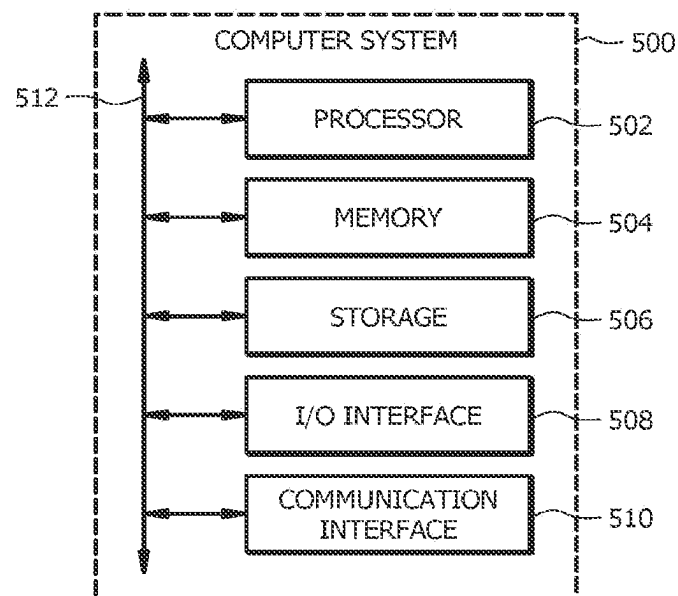


FIG. 4

*FIG. 5*

1

SYSTEMS AND METHODS FOR DETERMINING NETWORK COMPONENT SCORES USING BANDWIDTH CAPACITY

PRIORITY

This nonprovisional application is a continuation under 35 U.S.C. § 120 of U.S. patent application Ser. No. 17/857,861 filed on Jul. 5, 2022, which is a continuation, under 35 U.S.C. § 120 of U.S. patent application Ser. No. 16/696,203 filed on Nov. 26, 2019, and entitled “Systems and Methods for Determining Network Component Scores Using Bandwidth Capacity,” which claims priority to U.S. Provisional Patent Application No. 62/843,108 filed May 3, 2019, all of which are hereby incorporated by reference in their entirety.

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTORS

In accordance with 35 U.S.C. § 102 (b)(1) and to the Applicant’s knowledge, the following public disclosure has been made by inventors Satya Ranjan Mohanty, Mankamana Prasad Mishra, and Ali Sajassi, and/or by another who obtained the subject matter publicly disclosed directly or indirectly from the inventors, within one (1) year of the effective filing date of this application:

Mohanty, S. et al., Weighted HRW and its applications draft-mohanty-bess-weighted-hrw-00, <https://tools.ietf.org/html/draft-mohanty-bess-weighted-hrw-00>, dated Mar. 11, 2019, 11 pages.

TECHNICAL FIELD

This disclosure generally relates to determining network component scores, and more specifically to systems and methods for determining network component scores using bandwidth capacity.

BACKGROUND

In a networking environment, a plurality of objects may be mapped to a plurality of network components. The plurality of network components may change due to a network component either being removed from or added to the networking environment. As a result of this change, the objects may be remapped to the viable network components. This remapping of the objects may cause widespread disruption within the networking environment.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an example system for determining network component scores using bandwidth capacity;

FIG. 2 illustrates an example system for mapping objects to network components using network component scores;

FIG. 3 illustrates an example system for determining network component scores using normalized weights;

FIG. 4 illustrates an example method for determining network component scores using bandwidth capacity; and

FIG. 5 illustrates an example computer system that may be used by the systems and methods described herein.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Overview

According to an embodiment, a system includes one or more processors and one or more computer-readable non-

2

transitory storage media coupled to the one or more processors. The one or more computer-readable non-transitory storage media include instructions that, when executed by the one or more processors, cause one or more routers to perform operations including receiving data from a network component, determining a first link bandwidth capacity between a first router and a host device, and determining a first score for the first router based on the first link bandwidth capacity. The operations also include determining a second link bandwidth capacity between a second router and the host device and determining a second score for the second router based on the second link bandwidth capacity. The operations further include comparing at least the first score and the second score to determine a highest score and assigning an edge router associated with the highest score to communicate the data to the host device.

The operations may also include determining that the second link bandwidth capacity between the second router and the host device has changed from the second link bandwidth capacity to a revised second link bandwidth capacity. The operations may include revising the second score for the second router based on the revised second link bandwidth capacity to generate a revised second score for the second router. The operations may include comparing at least the first score and the revised second score to determine a highest score and reassigning the edge router associated with the highest score to communicate the first data to the host device.

Determining the first score for the first router based on the first link bandwidth capacity may include one or more of the following: determining a first router identifier for the first router; determining a data identifier for the data; calculating a normalized hash value of the first router identifier and the data identifier; calculating a natural logarithm of the normalized hash value; and dividing the natural logarithm of the normalized hash value into the first link bandwidth capacity to generate the first score (negating the weight or the final value by multiplying it by -1 , or using the absolute value in place of the negative elements where necessary to make the score positive). The data identifier may be associated with one of the following: an ethernet virtual private network (EVPN) instance (EVI) identifier or a bridge domain (BD) identifier. The edge router associated with the highest score may be a designated forwarder (DF) provider edge router. The data received from the network component may be associated with a multicast flow. The network component may be a router reflector for Border Gateway Protocol (BGP).

According to another embodiment, a method includes receiving, by a first router, data from a network component. The method also includes determining, by the first router, a first link bandwidth capacity between the first router and a host device and determining, by the first router, a first score for the first router based on the first link bandwidth capacity. The method also includes determining, by the first router, a second link bandwidth capacity between a second router and the host device and determining, by the first router, a second score for the second router based on the second link bandwidth capacity. The method further includes comparing, by the first router, at least the first score and the second score to determine a highest score and assigning, by the first router, an edge router associated with the highest score to communicate the data to the host device.

According to yet another embodiment, one or more computer-readable non-transitory storage media embody instructions that, when executed by a processor, cause the processor to perform operations including receiving data

from a network component, determining a first link bandwidth capacity between a first router and a host device, and determining a first score for the first router based on the first link bandwidth capacity. The operations also include determining a second link bandwidth capacity between a second router and the host device and determining a second score for the second router based on the second link bandwidth capacity. The operations further include comparing at least the first score and the second score to determine a highest score and assigning an edge router associated with the highest score to communicate the data to the host device.

Technical advantages of certain embodiments of this disclosure may include one or more of the following. Certain embodiments of this disclosure may reduce the computation of the network component scores from the product of the number of network components (e.g., the number of routers or servers) and the number of objects to just the number of objects (e.g., the number of flows). This reduction may result in significant savings in the context of virtual Ethernet Segment (ES) since several ESs may use the same physical link. Certain embodiments of this disclosure may reduce or eliminate disruption within a networking environment since very little movement of flows occurs between provider edges in the DF context. Certain embodiments of this disclosure provide a true Highest Random Weight (HRW) solution that is not stateful. As yet another advantage, embodiments of this disclosure may become more efficient as the number of VLANs, EVI identities, BD identities, and the like increase. Embodiments of this disclosure may solve the weighted DF load balancing problem in EVPN with minimal disruption. Certain embodiments described herein may be used for resilient hashing in unequal cost multipath (UCMP) load balancing in the forwarding information base (FIB).

Other technical advantages will be readily apparent to one skilled in the art from the following figures, descriptions, and claims. Moreover, while specific advantages have been enumerated above, various embodiments may include all, some, or none of the enumerated advantages.

Example Embodiments

Embodiments of this disclosure include systems and methods for determining network component scores using bandwidth capacity. Provider edge nodes of a network may receive data from a route reflector, and the provider edge node with the highest score is assigned to communicate the data to a customer edge node. The scores are determined using a weighted bandwidth context rather than a normalized weight associated with the provider edge node. As such, when a weight associated with a provider edge node (e.g., a router or server) changes, only the weight associated with the affected provider edge node needs to be computed. The weights associated with the other provider edge nodes do not need to be computed again.

In some embodiments, a method for designated forwarder (DF) election includes determining a weight of a server. The weight of the server is associated with a link bandwidth. The method for DF election also includes determining a normalized hash value using a server identification and an object identification and determining a natural logarithm of the normalized hash value. The method for DF election further includes determining a score for the server using the weight of the server, the normalized hash value, and the natural logarithm.

Instead of taking the normalized weight, the score is taken as: $-w_i/\ln(\text{hashn}(S_i, \text{Obj}))$, where: (i) "hashn" is the normalized hash (e.g., a value between 0 and 1 but not including

0); and (ii) "ln" is the natural logarithm of the normalized hash value. When w_i changes to w_i' , only the computation of objects with server S_i needs to be performed again. The other servers do not need to compute the weights again. An assumption in this technique is that the hash is a uniform hashing function, which is a requirement that is easily met in practice, especially if the object set is not small. Extrapolating this technique to the EVPN, the servers can be thought of as the PEs, and the key-ids as the EVI id or BD Id. DF Election is performed efficiently in a weighted bandwidth context. Good hash functions satisfy the uniform hashing properly. The object set is the number of VLANs/EVIs/BDs, which will be large in real deployments.

FIG. 1 shows an example system for determining network component scores using bandwidth capacity. FIG. 2 shows an example system for mapping objects to network components using network component scores, and FIG. 3 shows an example system for determining network component scores using normalized weights. FIG. 4 shows an example method for determining network component scores using bandwidth capacity. FIG. 5 shows an example computer system that may be used by the systems and methods described herein.

FIG. 1 illustrates an example system **100** for determining network component scores using bandwidth capacity. System **100** or portions thereof may be associated with an entity, which may include any entity, such as a business or company (e.g., a service provider) that determines network component scores using bandwidth capacity. The components of system **100** may include any suitable combination of hardware, firmware, and software. For example, the components of system **100** may use one or more elements of the computer system of FIG. 5.

System **100** includes a network **110**, a cloud environment **120**, a route reflector **130**, provider edge nodes **140**, a customer edge node **150**, and links **160**. Network **110** of system **100** is any type of network that facilitates communication between components of system **100**. Network **110** may connect one or more components of system **100**. This disclosure contemplates any suitable network. One or more portions of network **110** may include an ad-hoc network, an intranet, an extranet, a VPN (e.g., an Ethernet VPN (EVPN)), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a cellular telephone network, a combination of two or more of these, or other suitable types of networks. Network **110** may include one or more networks. Network **110** may be any communications network, such as a private network, a public network, a connection through Internet, a mobile network, a WI-FI network, etc. One or more components of system **100** may communicate over network **110**. Network **110** may include a core network (e.g., the Internet), an access network of a service provider, an Internet service provider (ISP) network, and the like. In certain embodiments, one or more portions of network **110** may utilize Multiprotocol Label Switching (MPLS). For example, one or more portions of network **110** may be an MPLS VPN. Network **110** of system **100** may include one or more cloud networks.

Cloud environment **120** of system **100** is a network environment that provides on-demand computer system resources (e.g., data storage and computing power) without direct active management by a user. Cloud environment **120** may include one or more data centers available to many users over the Internet. Cloud environment **120** may be limited to a single organization (e.g., an enterprise cloud

environment) or be available to many organizations (e.g., a public cloud environment). In certain embodiments, cloud environment **120** is an internal BGP (iBGP) cloud environment. Cloud environment **120** of FIG. **1** includes route reflector **130**.

Route reflector **130** of system **100** is a network routing component for BGP. Route reflector **130** may be a router that acts as a routing information exchange server for other iBGP routers. Route reflector **130** may advertise routes between internal BGP peers (e.g., provider edge nodes **140**). A new node (e.g., provider edge node **140**) in network **110** may receive all routes from route reflector **130** in response to creating a session with route reflector **130**.

Provider edge nodes **140** of system **100** are connection points within network **110** that receive, create, store and/or send data along a path. Provider edge nodes **140** may include one or more redistribution points that recognize, process, and forward data to other nodes (e.g., customer edge node **150**) of network **110**. Provider edge nodes **140** may include virtual and/or physical provider edge nodes. In certain embodiments, one or more provider edge nodes **140** include data communications equipment such as switches, bridges, modems, hubs, and the like. In some embodiments, one or more provider edge nodes **140** include data terminal equipment such as routers, servers, printers, workstations, and the like. In certain embodiments, one or more provider edge nodes **140** are DFs. In the illustrated embodiment of FIG. **1**, provider edge nodes **140** include provider edge node **140a**, provider edge node **140b**, and provider edge node **140c**.

In certain embodiments, provider edge nodes **140** may be part of a multi-homing configuration. For example, provider edge nodes **140** may be part of the same multi-homing redundancy group. Provider edge nodes **140** may operate in all-active mode. Each provider edge node **140** may include a group of member ports. Each member port may have a different bandwidth capacity. Each provider edge node **140** of system **100** in a redundancy group may be provisioned with a group of member ports with different bandwidth capacities. For example, provider edge node **140a** may include a first port having a first bandwidth capacity, a second port having a second bandwidth capacity, and a third port having a third bandwidth capacity.

FIG. **1** may be described using the following three cases: (1) in the case of link bandwidth provisioning distribution, some deployment may exist where each PE in a redundancy group has been provisioned with a group of member ports with different capacity; (2) in the case of EVI bandwidth utilization policy, a bundle (e.g., Green Bundle) that connects the CE to the PEs denotes one of the Ethernet Segments(ES). These ESs may be shared among different sets of EVIs. A user may wish to gain more control by restricting the amount of bandwidth each EVI can use. For example, if the ES has a total of 100 gigabytes of capacity, the user may wish to distribute available bandwidth across multiple EVIs (e.g., EVI-1, EVI-2, and EVI-3 in ratio of 10 gigabytes, 50 Gigabytes, and 40 Gigabytes, respectively); (3) in the case of per multicast flow bandwidth requirement, there are certain use cases where per multicast flow, bandwidth requirements are pre-defined.

Request for Comments (RFC) 7432, Border Gateway Protocol (BGP) Multiprotocol Label Switching (MPLS)-Based Ethernet virtual private network (VPN), defines the standard to support multi-homing in the Ethernet VPN (EVPN). RFC 7432 defines the procedures for EVPN designated forwarder (DF) election in cases of all-active and single active mode. This base standard defines procedures to elect EVPN DF where granularity for DF election is defined

per Ethernet Segment(ES) per Ethernet Virtual Interconnect (EVI). RFC 8584, Framework for Ethernet VPN Designated Forwarder Election Extensibility, improves upon RFC 7432 by introducing the Highest Random Weight (HRW) framework that provides for better carving. RFC 7432 provides for better carving at a further granularity of per ES per BD. Per multicast flow, DF election provides a solution for performing the DF election for multicast in the EVPN.

In some embodiments, a bundle (e.g., an ES) may connect each provider edge node **140** to customer edge node **150**. These ESs may be shared among different sets of EVIs. One or more provider edge nodes **140** may restrict the amount of bandwidth each EVI can use. For example, if an ES has a total of 100 gigabytes of capacity, provider edge node **140a** of system **100** may distribute available bandwidth across EVI-1, EVI-2, and EVI-3 in ratio of 10 gigabytes, 50 gigabytes, and 40 gigabytes, respectively. In certain embodiments, data received by one or more provider edge nodes **140** may be associated with a multicast flow. The bandwidth requirements may be pre-defined per multicast flow. For example, the bandwidth requirements may be pre-defined for the dissemination of raw video. The data may have a constant bit rate and/or a pre-defined associated bandwidth requirement.

One or more provider edge nodes **140** may determine which provider edge node **140** should route the incoming traffic based on the link bandwidth capacity associated with each provider edge node **140**. For example, provider edge node **140a** may determine a bandwidth capacity of link **160a** between provider edge node **140a** and customer edge node **150**. Provider edge node **140a** may determine a score for provider edge node **140a** based on the bandwidth capacity of link **160a**. Provider edge node **140a** may calculate the score based on an identifier of provider edge node **140a**, an identifier (e.g., an EVI identifier or a BD identifier) for the incoming data to be assigned, and the bandwidth capacity for link **160a**. For example, provider edge node **140a** may calculate a normalized hash value of the identifier for provider edge node **140a** and the data identifier, calculate a natural logarithm of the normalized hash value, and divide the natural logarithm of the normalized hash value into the bandwidth capacity of link **160a** to generate the score for provider edge node **140a** (negating or using the absolute value of any negative elements where necessary to make the score positive).

Similarly, provider edge node **140a** may determine a bandwidth capacity of link **160b** between provider edge node **140b** and customer edge node **150**. Provider edge node **140a** may determine a score for provider edge node **140b** based on the bandwidth capacity of link **160b**. Provider edge node **140a** may calculate the score based on an identifier of provider edge node **140b**, an identifier for the incoming data to be assigned, and the bandwidth capacity of link **160b**. For example, provider edge node **140a** may calculate a normalized hash value of the identifier for provider edge node **140b** and the data identifier, calculate a natural logarithm of the normalized hash value, and divide the natural logarithm of the normalized hash value into the bandwidth capacity of link **160b** to generate the score for provider edge node **140b** (negating or using the absolute value of any negative elements where necessary to make the score positive).

In a like manner, provider edge node **140a** may determine a bandwidth capacity of link **160c** between provider edge node **140c** and customer edge node **150**. Provider edge node **140a** may determine a score for provider edge node **140c** based on the bandwidth capacity of link **160c**. Provider edge node **140a** may calculate the score based on an identifier of

provider edge node **140c**, an identifier for the incoming data to be assigned, and the bandwidth capacity of link **160c**. For example, provider edge node **140a** may calculate a normalized hash value of the identifier for provider edge node **140c** and the data identifier, calculate a natural logarithm of the normalized hash value, and divide the natural logarithm of the normalized hash value into the bandwidth capacity of link **160c** to generate the score for provider edge node **140c** (negating or using the absolute value of any negative elements where necessary to make the score positive).

Provider edge node **140a** may compare the scores for provider edge nodes **140a**, **140b**, and **140c** to determine which provider edge node **140** has the highest score. Provider edge node **140a** may assign the node (e.g., provider edge node **140a**, **140b**, or **140c**) with the highest score to communicate the data to customer edge node **150**. For example, if provider edge node **140a** determines that the score for provider edge node **140b** is higher than each of the scores for provider edge nodes **140a** and **140c**, provider edge node **140a** may assign the data to provider edge node **140b**. Provider edge node **140b** then communicates the data to customer edge node **150** along link **160b**.

If multiple provider edge nodes **140** have the same highest score, provider edge node **140** with the highest IP address is preferred. For example, if the scores for provider edge nodes **140a** and **140b** are equal, the scores for provider edge nodes **140a** and **140b** are higher than the score for provider edge node **140c**, and the IP address of provider edge node **140a** is higher than the IP address for provider edge node **140b**, then provider edge node **140a** assigns the data to itself. Provider edge node **140a** then communicates the data to customer edge node **150** along link **160a**.

In certain embodiments, the bandwidth capacity of one or more links **160** may change. For example, the bandwidth capacity of link **160a** may be eliminated if provider edge node **140a** is reprovisioned or inactivated. As another example, the available bandwidth capacity of link **160a** may increase or decrease in accordance with one or more policies. If the bandwidth capacity of one or more links **160** changes, one or more provider edge nodes **140** may revise the score for provider edge node **140** associated with affected link **160**. For example, if the bandwidth capacity of link **160a** between provider edge node **140a** and customer edge node **150** increases, only the scores for the data assigned to provider edge node **140a** are recalculated. The scores for the data assigned to provider edge nodes **140b** and **140c** do not need to be recalculated.

In certain embodiments, protocol extensions are defined so that all provider edge nodes **140** are aware that assignments are determined in a weighted bandwidth context. For example, for DF election, each provider edge node **140** may be aware that the DF Election they are participating in is with respect to a weighted HRW. The property of consistent hashing/rendezvous hashing is retained.

Customer edge node **150** of system **100** is a connection point within network **110** that receives, creates, stores, and/or sends data. Customer edge node **150** may include one or more endpoints and/or one or more redistribution points that recognize data, process data, and/or forward data to other nodes of network **110**. Customer edge node **150** may include virtual and/or physical customer edge nodes. In certain embodiments, customer edge node **150** includes data communications equipment such as switches, bridges, modems, hubs, and the like. In some embodiments, customer edge node **150** includes data terminal equipment such as routers, servers, printers, workstations, and the like. In certain embodiments, customer edge node **150** is a host

device. Customer edge node **150** may receive incoming traffic from one or more provider edge nodes **140**.

Links **160** (e.g., link **160a**, link **160b**, and link **160c**) of system **100** are communication channels within network **110**. Each link **160** connects two nodes of network **110**. One or more links **160** of network **110** may be wired or wireless. One or more links **160** may include Ethernet links, ESs, and the like. Each link **160** has a bandwidth capacity. The bandwidth capacity of each link **160** is the capacity of the respective link to transmit data (e.g., a maximum amount of data) from one node (e.g., provider edge node **140a**, **140b**, or **140c**) of network **110** to another node (e.g., customer edge node **150**) of network **110** in a given amount of time (e.g., one second). Link **160a** connects provider edge node **140a** to customer edge node **150**, link **160b** connects provider edge node **140b** to customer edge node **150**, and link **160c** connects provider edge node **140c** to customer edge node **150**. The bandwidth capacity of links **160a**, **160b**, and **160c** of FIG. 1 is represented by the line thickness. Link **160a** has a higher bandwidth capacity (which is represented by a thicker line) than links **160b** and **160c**. Link **160c** has a higher bandwidth capacity than link **160b** but a lower bandwidth capacity than link **160a**.

In operation, provider edge node **140a** of system **100** receives data from route reflector **130** of cloud environment **120**. Provider edge node **140a** determines a bandwidth capacity of link **160a** between provider edge node **140a** and customer edge node **150**. Provider edge node **140a** determines a score for provider edge node **140a** by calculating a normalized hash value of the identifier for provider edge node **140a** and the data identifier (e.g., an EVI identifier or a BD identifier), calculating a natural logarithm of the normalized hash value, and dividing the natural logarithm of the normalized hash value into the bandwidth capacity (negating or using the absolute value of any negative elements where necessary to make the score positive). Similarly, provider edge node **140a** determines a score provider edge node **140b** and provider edge node **140c** based on the bandwidth capacities of links **160b** and **160c**, respectively. Provider edge node **140a** then assigns the provider edge node with the highest score to communicate the data to customer edge node **150**. As such, system **100** may be used to determine network component scores based on normalized hash values rather than normalized weights, which may reduce or eliminate disruptive remapping of data to provider edge nodes **140**.

Although FIG. 1 illustrates a particular arrangement of network **110**, cloud environment **120**, route reflector **130**, provider edge nodes **140**, customer edge node **150**, and links **160**, this disclosure contemplates any suitable arrangement of network **110**, cloud environment **120**, route reflector **130**, provider edge nodes **140**, customer edge node **150**, and links **160**. For example, route reflector **130** of system **100** may not be located within cloud environment **120**.

Although FIG. 1 illustrates a particular number of networks **110**, cloud environments **120**, route reflectors **130**, provider edge nodes **140**, customer edge nodes **150**, and links **160**, this disclosure contemplates any suitable number of networks **110**, cloud environments **120**, route reflectors **130**, provider edge nodes **140**, customer edge nodes **150**, and links **160**. For example, system **100** may include more or less than three provider edge nodes **140**.

Although FIG. 1 describes and illustrates particular components performing particular actions of system **100**, this disclosure contemplates any suitable combination of any suitable components, devices, or systems carrying out any

suitable actions of system **100**. For example, a network controller may perform one or more of the calculations described in FIG. 1.

FIG. 2 illustrates an example system for mapping objects to network components using network component scores. FIG. 2 includes the following network components: node N1, node N2, and node N3. Each node may represent a router, a server, and the like. In certain embodiments, node N1 may represent provider edge node **140a** of FIG. 1, node N2 may represent provider edge node **140b** of FIG. 1, and node N3 may represent provider edge node **140c** of FIG. 1. The objects of FIG. 2 may represent the data received by provider edge nodes **140** of FIG. 1. Several nodes (e.g., nodes N1, N2, and N3) may have the ability to each process the same object.

FIG. 2 includes the following objects: object **233**, object **318**, and object **597**. The modulo-N algorithm (i.e., the object identifier divided by the number of nodes) may not be suitable for mapping objects to network components. For example, when a node is removed or added (i.e., the number of nodes changes), disruptive remapping of the objects to the nodes may occur. Objects that were previously mapped to nodes that were not removed or added are often re-assigned. Two algorithms that solve this problem include consistent hashing and rendezvous hashing (e.g., HRW). In rendezvous hashing, a hash is computed from the concatenation of the node identifier (e.g., the node-id) and the object (e.g., the key). If the *i*th node is denoted by *N_i*, then the score for object *Obj* on node *N_i* is defined to be: $\text{Score}(\text{Obj}, N_i) = \text{Hash}(\text{Obj} * N_i)$, where “*” denotes the concatenation. The designated node for object *Obj* is the node *N_j* for which $\text{Score}(\text{Obj}, N_j) \geq \text{Score}(\text{Obj}, N_i)$ for all *i*=1, *n*, where *n* represents any suitable integer. The node for which the hash value is the highest wins and is considered as the designated node for that object. In the case of a tie, the node with the highest IP address is preferred.

In the illustrated embodiment of FIG. 2, the scores for object **233** are calculated as follows: the hash of node N1 and object **233** (i.e., $H(N1 * 233)$) equals score 457; the hash of node N2 and object **233** (i.e., $H(N2 * 233)$) equals score 317, and the hash of node N3 and object **233** (i.e., $H(N3 * 233)$) equals score 512. Since 512 is the highest of the three scores, object **233** is assigned to node N3. Similarly, the scores for object **318** are calculated as follows: the hash of node N1 and object **318** (i.e., $H(N1 * 318)$) equals score 471; the hash of node N2 and object **318** (i.e., $H(N2 * 318)$) equals score 513, and the hash of node N3 and object **318** (i.e., $H(N3 * 318)$) equals score 172. Since 513 is the highest of the three scores, object **318** is assigned to node N2. Likewise, the scores for object **597** are calculated as follows: the hash of node N1 and object **597** (i.e., $H(N1 * 597)$) equals score 919; the hash of node N2 and object **597** (i.e., $H(N2 * 597)$) equals score 200, and the hash of node N3 and object **597** (i.e., $H(N3 * 597)$) equals score 706. Since 919 is the highest of the three scores, object **597** is assigned to node N1. As such, the objects are assigned to nodes based on network component scores calculated using HRW.

If node *N_j* goes down, then only those objects for which node *N_j* is the designated node need to be remapped. Other objects remain unaffected. Similarly, when a new node *N_k* comes up, node *N_k* will be the designated node for only those objects for which its hash with them is the highest. Rendezvous hashing may be applied in the load balancing context and also in many networking applications. Rendezvous hashing may also be used in the context of EVPN in the HRW DF Election, multicast, and port active applications. The underlying assumption for HRW is that all nodes are of

the same capacities or “weight.” When this assumption is not valid, the standard HRW algorithm may not be sufficient, as it will not result in a distribution of objects to nodes in the ratio of their weights.

FIG. 3 illustrates an example system **300** for determining network component scores using normalized weights. FIG. 3 includes the following network components: node N1, node N2, node N3, and node N4. Each node may represent a server, a router, and the like. Several nodes (e.g., nodes N1, N2, N3, and N4) can each process the same object. A weight is assigned to each node, and a normalized weighted hash is taken as the score. The four nodes N1, N2, N3, and N4 of FIG. 3, respectively, have the following different capacities (*W_i*): 50, 15, 20 and 15. An object with key-id **233** is to be assigned to a node. The nodes are assigned normalized weights as per their capacities. Node *N_i* is assigned a weight $W_i / \text{sum}(W_i)$, where $\text{sum}(W_i)$ is defined to be the sum of all the weights. In the illustrated embodiment of FIG. 3, $\text{sum}(W_i)$ is 100. Accordingly, the weighted hash values are taken, and the object is assigned to node N1 since that weighted hash value associated with node N1 is the highest. This same process may be repeated for all the objects (e.g., objects **318** and **597** of FIG. 2).

In the illustrated embodiment of FIG. 3, the scores for object **233** are calculated as follows: the hash of node N1 and object **233** times weight *W₁* (i.e., $(H(N1 * 233) * 0.50)$) equals score 228.5; the hash of node N2 and object **233** times weight *W₂* (i.e., $(H(N2 * 233) * 0.15)$) equals score 47.55; the hash of node N3 and object **233** times weight *W₃* (i.e., $(H(N3 * 233) * 0.20)$) equals score 102.4; and the hash of node N4 and object **233** times weight *W₄* ($(H(N4 * 233) * 0.15)$) equals score 35.4. Since 228.5 is the highest of the four scores, object **233** is assigned to node N1. Similar computations may be applied to other objects (e.g., objects **318** and **597** of FIGURE 2).

In certain embodiments, the weight of a node may change. For example, the weight *W₂* of N2 may change from 15 to 25. Accordingly, the normalized value of each of the nodes (i.e., nodes N1, N2, N3, and N4) changes, and the normalized hash value is recomputed for each node. The process is repeated for each object. Because of this re-computation of every object with every node, movement of objects (e.g., Virtual Local Area Networks (VLANs)) across nodes (e.g., provider edge nodes) will likely occur.

FIG. 4 illustrates an example method **400** for determining network component scores using bandwidth capacity. Rather than using the normalized weight of the network component as illustrated in FIG. 3 above, the score is taken as: $W_i / \ln(\text{hashn}(N_i, \text{Obj}))$, where: *W_i* represents the weight of the network component; *N_i* represents the identity of the network component; hashn represents the normalized hash value (e.g., a value between 0 and 1 but not including 0); and $\ln$ represents the natural logarithm of the normalized hash value. When the hash value is normalized to the range between 0 and 1, the logarithm of the value is negative. In this case the absolute value is used for the score or the value of the equation is negated (e.g., $-W_i / \ln(\text{hashn}(N_i, \text{Obj}))$). Accordingly, when *W_i* changes to *W_i'*, only the computation of objects with node *N_i* needs to be performed again. The other nodes do not need to compute the weights again. An assumption in this method is that the hash is a uniform hashing function. Extrapolating this technique to the EVPN, the nodes/servers are provider edge (PE) nodes, and the object identifiers/key-ids are the EVI identities or BD identities. DF Election is performed efficiently in a weighted bandwidth context. The object set may be the number of VLANs, EVIs, BDs, and the like, which may be large.

11

In case there are other types of traffic like unicast and aliased traffic, the user may wish to exclude that bandwidth while doing the EVI/(S/G) carving in the weighted link bandwidth case. It would practically not be possible to keep polling existing link uses. Per (ES, EVI) periodic link use will need to be calculated, which will be communicated to peers only once it reaches certain threshold. For example, if user policy defines four windows to cover link uses, then reporting is only done when overall uses spill over from one window to other. For example, initially, if link uses are <25%, it would be reported once, and the next update would be only when bandwidth uses go beyond 25%.

The weight (W_i) of the network component is the bandwidth capacity of the link associated with the network component. For example, referring to FIG. 1, the weight W_i of provider edge node 140a is the bandwidth capacity of link 160a, the weight W_i of provider edge node 140b is the bandwidth capacity of link 160b, and the weight W_i of provider edge node 140c is the bandwidth capacity of link 160c. As such, assignments (e.g., DF election) of network components are determined in a weighted bandwidth context.

Method 400 begins at step 405. At step 410, a network node (e.g., provider edge node 140a of FIG. 1) of a network (e.g., network 110 of FIG. 1) receives data (e.g., an object) from a network component (e.g., route reflector 130 of FIG. 1). Method 400 then moves from step 410 to step 415, where the network node determines a first link bandwidth capacity between a first router (e.g., provider edge node 140a of FIG. 1) and a host device (e.g., customer edge node 150 of FIG. 1). In certain embodiments, the network node and the first router are the same network component. Method 400 then moves from step 415 to step 420.

At step 420, the network node determines a first score for the first router based on the first link bandwidth capacity. For example, the network node may determine a first router identifier for the first router and a data identifier for the data. In certain embodiments, the data identifier may be an EVI identifier or a BD identifier. The network node may use the first router identifier and the data identifier to calculate a normalized hash value of the first router identifier and the data identifier. The network node may calculate a natural logarithm of the normalized hash value and divide the natural logarithm of the normalized hash value into the first link bandwidth capacity to generate the first score (negating or using the absolute value of any negative elements where necessary to make the score positive). Method 400 then moves from step 420 to step 425.

At step 425, the network node determines a second link bandwidth capacity between a second router (e.g., provider edge node 140b of FIG. 1) and the host device. Method 400 then moves from step 425 to step 430, where the network node determines a second score for the second router based on the second link bandwidth capacity. For example, the network node may determine a second router identifier for the second router. The network node may use the second router identifier and the data identifier to calculate a normalized hash value of the second router identifier and the data identifier. The network node may calculate a natural logarithm of the normalized hash value and divide the natural logarithm of the normalized hash value into the second link bandwidth capacity to generate the second score (negating or using the absolute value of any negative elements where necessary to make the score positive). Method 400 then moves from step 430 to step 435.

At step 435, the network node determines whether the first score is different than the second score. If the first score is

12

not different than the second score, method 400 moves from step 435 to step 440, where the network node determines whether the IP address of the first router is higher than the IP address of the second router. If the IP address of the first router is higher than the IP address of the second router, method 400 moves from step 440 to step 450, where the network node assigns the first router to communicate the data to the host device. If, at step 440, the network node determines that the IP address of the first router is not higher than the IP address of the second router, method 400 moves from step 440 to step 455, where the network node assigns the second router to communicate the data to the host device.

If, at step 435, the network node determines that the first score is different than the second score, method 400 moves from step 435 to step 445, where the network node determines whether the first score associated with the first router is higher than the second score associated with the second router. If the first score is higher than the second score, method 400 moves from step 445 to step 450, where the network node assigns the first router to communicate the data to the host device. If, at step 440, the network node determines that the first score is not higher than the second score, method 400 moves from step 445 to step 455, where the network node assigns the second router to communicate the data to the host device. Method 400 then moves from steps 450 and 455 to step 460, where method 400 ends. As such, method 400 determines network component scores based on bandwidth capacity using normalized hash values rather than normalized weights, which may reduce or eliminate disruptive remapping of data to routers.

Method 400 may reduce or eliminate the disruptive mapping of objects to nodes (e.g., servers). If the weight (represented as the bandwidth of link 160b of FIG. 1) of a node (e.g., provider edge node 140b of FIG. 1) changes, then the computation in the steps above pertaining to the logarithm are recalculated only for the objects (e.g., VLANs) that were previously assigned only to the node. Other objects that were previously assigned to other nodes (e.g., provider edge nodes 140a and 140c of FIG. 1) continue to be assigned as before. As such, method 400 reduces the assignment problem from a complexity of $N*O$ (i.e., number of nodes $\times$ number of objects) to just O (i.e., number of objects).

Although this disclosure describes and illustrates particular steps of method 400 of FIG. 4 as occurring in a particular order, this disclosure contemplates any suitable steps of method 400 of FIG. 4 occurring in any suitable order. For example, step 410 directed to receiving data from a network component may be performed after steps 415 and 425 directed to determining link bandwidth capacities.

Although this disclosure describes and illustrates an example method 400 for determining network component scores using bandwidth capacity including the particular steps of the method of FIG. 4, this disclosure contemplates any suitable method 400 for determining network component scores using bandwidth capacity, including any suitable steps, which may include all, some, or none of the steps of the method of FIG. 4, where appropriate. For example, method 400 may include additional steps directed to determining a third link bandwidth capacity between a third router (e.g., provider edge node 140c of FIG. 1) and the host device, determining a third score for the third router based on the third link bandwidth capacity, determining the highest score, and assigning the router with the highest score to communicate the data to the host device.

Although this disclosure describes and illustrates particular components, devices, or systems carrying out particular steps of method 400 of FIG. 4, this disclosure contemplates

13

any suitable combination of any suitable components, devices, or systems carrying out any suitable steps of method **400** of FIG. **4**.

FIG. **5** illustrates an example computer system **500**. In particular embodiments, one or more computer systems **500** perform one or more steps of one or more methods described or illustrated herein. In particular embodiments, one or more computer systems **500** provide functionality described or illustrated herein. In particular embodiments, software running on one or more computer systems **500** performs one or more steps of one or more methods described or illustrated herein or provides functionality described or illustrated herein. Particular embodiments include one or more portions of one or more computer systems **500**. Herein, reference to a computer system may encompass a computing device, and vice versa, where appropriate. Moreover, reference to a computer system may encompass one or more computer systems, where appropriate.

This disclosure contemplates any suitable number of computer systems **500**. This disclosure contemplates computer system **500** taking any suitable physical form. As example and not by way of limitation, computer system **500** may be an embedded computer system, a system-on-chip (SOC), a single-board computer system (SBC) (such as, for example, a computer-on-module (COM) or system-on-module (SOM)), a desktop computer system, a laptop or notebook computer system, an interactive kiosk, a mainframe, a mesh of computer systems, a mobile telephone, a personal digital assistant (PDA), a server, a tablet computer system, an augmented/virtual reality device, or a combination of two or more of these. Where appropriate, computer system **500** may include one or more computer systems **500**; be unitary or distributed; span multiple locations; span multiple machines; span multiple data centers; or reside in a cloud, which may include one or more cloud components in one or more networks. Where appropriate, one or more computer systems **500** may perform without substantial spatial or temporal limitation one or more steps of one or more methods described or illustrated herein. As an example and not by way of limitation, one or more computer systems **500** may perform in real time or in batch mode one or more steps of one or more methods described or illustrated herein. One or more computer systems **500** may perform at different times or at different locations one or more steps of one or more methods described or illustrated herein, where appropriate.

In particular embodiments, computer system **500** includes a processor **502**, memory **504**, storage **506**, an input/output (I/O) interface **508**, a communication interface **510**, and a bus **512**. Although this disclosure describes and illustrates a particular computer system having a particular number of particular components in a particular arrangement, this disclosure contemplates any suitable computer system having any suitable number of any suitable components in any suitable arrangement.

In particular embodiments, processor **502** includes hardware for executing instructions, such as those making up a computer program. As an example and not by way of limitation, to execute instructions, processor **502** may retrieve (or fetch) the instructions from an internal register, an internal cache, memory **504**, or storage **506**; decode and execute them; and then write one or more results to an internal register, an internal cache, memory **504**, or storage **506**. In particular embodiments, processor **502** may include one or more internal caches for data, instructions, or addresses. This disclosure contemplates processor **502** including any suitable number of any suitable internal

14

caches, where appropriate. As an example and not by way of limitation, processor **502** may include one or more instruction caches, one or more data caches, and one or more translation lookaside buffers (TLBs). Instructions in the instruction caches may be copies of instructions in memory **504** or storage **506**, and the instruction caches may speed up retrieval of those instructions by processor **502**. Data in the data caches may be copies of data in memory **504** or storage **506** for instructions executing at processor **502** to operate on; the results of previous instructions executed at processor **502** for access by subsequent instructions executing at processor **502** or for writing to memory **504** or storage **506**; or other suitable data. The data caches may speed up read or write operations by processor **502**. The TLBs may speed up virtual-address translation for processor **502**. In particular embodiments, processor **502** may include one or more internal registers for data, instructions, or addresses. This disclosure contemplates processor **502** including any suitable number of any suitable internal registers, where appropriate. Where appropriate, processor **502** may include one or more arithmetic logic units (ALUs); be a multi-core processor; or include one or more processors **502**. Although this disclosure describes and illustrates a particular processor, this disclosure contemplates any suitable processor.

In particular embodiments, memory **504** includes main memory for storing instructions for processor **502** to execute or data for processor **502** to operate on. As an example and not by way of limitation, computer system **500** may load instructions from storage **506** or another source (such as, for example, another computer system **500**) to memory **504**. Processor **502** may then load the instructions from memory **504** to an internal register or internal cache. To execute the instructions, processor **502** may retrieve the instructions from the internal register or internal cache and decode them. During or after execution of the instructions, processor **502** may write one or more results (which may be intermediate or final results) to the internal register or internal cache. Processor **502** may then write one or more of those results to memory **504**. In particular embodiments, processor **502** executes only instructions in one or more internal registers or internal caches or in memory **504** (as opposed to storage **506** or elsewhere) and operates only on data in one or more internal registers or internal caches or in memory **504** (as opposed to storage **506** or elsewhere). One or more memory buses (which may each include an address bus and a data bus) may couple processor **502** to memory **504**. Bus **512** may include one or more memory buses, as described below. In particular embodiments, one or more memory management units (MMUs) reside between processor **502** and memory **504** and facilitate accesses to memory **504** requested by processor **502**. In particular embodiments, memory **504** includes random access memory (RAM). This RAM may be volatile memory, where appropriate. Where appropriate, this RAM may be dynamic RAM (DRAM) or static RAM (SRAM). Moreover, where appropriate, this RAM may be single-ported or multi-ported RAM. This disclosure contemplates any suitable RAM. Memory **504** may include one or more memories **504**, where appropriate. Although this disclosure describes and illustrates particular memory, this disclosure contemplates any suitable memory.

In particular embodiments, storage **506** includes mass storage for data or instructions. As an example and not by way of limitation, storage **506** may include a hard disk drive (HDD), a floppy disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus (USB) drive or a combination of two or more of these. Storage **506** may include removable or non-removable (or

15

fixed) media, where appropriate. Storage **506** may be internal or external to computer system **500**, where appropriate. In particular embodiments, storage **506** is non-volatile, solid-state memory. In particular embodiments, storage **506** includes read-only memory (ROM). Where appropriate, this ROM may be mask-programmed ROM, programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), electrically alterable ROM (EAROM), or flash memory or a combination of two or more of these. This disclosure contemplates mass storage **506** taking any suitable physical form. Storage **506** may include one or more storage control units facilitating communication between processor **502** and storage **506**, where appropriate. Where appropriate, storage **506** may include one or more storages **506**. Although this disclosure describes and illustrates particular storage, this disclosure contemplates any suitable storage.

In particular embodiments, I/O interface **508** includes hardware, software, or both, providing one or more interfaces for communication between computer system **500** and one or more I/O devices. Computer system **500** may include one or more of these I/O devices, where appropriate. One or more of these I/O devices may enable communication between a person and computer system **500**. As an example and not by way of limitation, an I/O device may include a keyboard, keypad, microphone, monitor, mouse, printer, scanner, speaker, still camera, stylus, tablet, touch screen, trackball, video camera, another suitable I/O device or a combination of two or more of these. An I/O device may include one or more sensors. This disclosure contemplates any suitable I/O devices and any suitable I/O interfaces **508** for them. Where appropriate, I/O interface **508** may include one or more device or software drivers enabling processor **502** to drive one or more of these I/O devices. I/O interface **508** may include one or more I/O interfaces **508**, where appropriate. Although this disclosure describes and illustrates a particular I/O interface, this disclosure contemplates any suitable I/O interface.

In particular embodiments, communication interface **510** includes hardware, software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) between computer system **500** and one or more other computer systems **500** or one or more networks. As an example and not by way of limitation, communication interface **510** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network or a wireless NIC (WNIC) or wireless adapter for communicating with a wireless network, such as a WI-FI network. This disclosure contemplates any suitable network and any suitable communication interface **510** for it. As an example and not by way of limitation, computer system **500** may communicate with an ad hoc network, a personal area network (PAN), a LAN, a WAN, a MAN, or one or more portions of the Internet or a combination of two or more of these. One or more portions of one or more of these networks may be wired or wireless. As an example, computer system **500** may communicate with a wireless PAN (WPAN) (such as, for example, a BLUETOOTH WPAN), a WI-FI network, a WI-MAX network, a cellular telephone network (such as, for example, a Global System for Mobile Communications (GSM) network, a Long-Term Evolution (LTE) network, or a 5G network), or other suitable wireless network or a combination of two or more of these. Computer system **500** may include any suitable communication interface **510** for any of these networks, where appropriate. Communication interface **510** may include one or more communication

16

interfaces **510**, where appropriate. Although this disclosure describes and illustrates a particular communication interface, this disclosure contemplates any suitable communication interface.

In particular embodiments, bus **512** includes hardware, software, or both coupling components of computer system **500** to each other. As an example and not by way of limitation, bus **512** may include an Accelerated Graphics Port (AGP) or other graphics bus, an Enhanced Industry Standard Architecture (EISA) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an Industry Standard Architecture (ISA) bus, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCIe) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local (VLB) bus, or another suitable bus or a combination of two or more of these. Bus **512** may include one or more buses **512**, where appropriate. Although this disclosure describes and illustrates a particular bus, this disclosure contemplates any suitable bus or interconnect.

Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such, as for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs), hybrid hard drives (HHDs), optical discs, optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, solid-state drives (SSDs), RAM-drives, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, non-volatile, or a combination of volatile and non-volatile, where appropriate.

Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated otherwise by context.

Embodiments of the present disclosure may provide certain advantages. As discussed above, embodiments of this disclosure may reduce the computation from the product of `#num_servers` and `#num_objects` to just the `#num_objects`. In the case of virtual ES, the savings may be significant because several ESs can use the same physical link. The assumption on the hash are met with any good standard hash function. As another advantage, certain embodiments of this disclosure solve the important property of disruption, so there will be almost very little movement of veins between PEs for the DF role. The carving is much better. As still another advantage, embodiments of this disclosure provide a true HRW solution that is not stateful. As yet another advantage, embodiments of this disclosure may become more efficient as the number of VLANs/EVI ids/BD ids increase, which holds for realistic deployments. Embodiments of this disclosure solve the weighted DF load balancing problem in EVPN with minimal disruption. The embodiments described herein may be used for resilient hashing in unequal cost multipath (UCMP) load balancing in the forwarding information base (FIB).

17

The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example 5 embodiments described or illustrated herein. Moreover, although this disclosure describes and illustrates respective embodiments herein as including particular components, elements, feature, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, features, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative. Additionally, although this disclosure describes or illustrates particular 20 embodiments as providing particular advantages, particular embodiments may provide none, some, or all of these advantages. The embodiments disclosed herein are only examples, and the scope of this disclosure is not limited to them. Particular embodiments may include all, some, or none of the components, elements, features, functions, operations, or steps of the embodiments disclosed herein. Embodiments according to the disclosure are in particular disclosed in the attached claims directed to a method, a storage medium, a system and a computer program product, wherein any feature mentioned in one claim category, e.g. method, can be claimed in another claim category, e.g. system, as well. The dependencies or references back in the attached claims are chosen for formal reasons only. However, any subject matter resulting from a deliberate reference back to any previous claims (in particular multiple dependencies) can be claimed as well, so that any combination of claims and the features thereof are disclosed and can be claimed regardless of the dependencies chosen in the attached claims. The subject-matter which can be claimed comprises not only the combinations of features as set out in the attached claims but also any other combination of features in the claims, wherein each feature mentioned in the claims can be combined with any other feature or combination of other features in the claims. Furthermore, any of the embodiments and features described or depicted herein can be claimed in a separate claim and/or in any combination 50 with any embodiment or feature described or depicted herein or with any of the features of the attached claims.

What is claimed is:

1. A method, comprising: 55
 at a first provider edge (PE) device in a multihoming group of PE devices in an Ethernet virtual private network (EVPN), the group comprising the first PE device and a second PE device, performing a designated forwarder (DF) election procedure including: 60
 receiving a request to route incoming traffic through one or more links within the EVPN;
 determining that the incoming traffic comprises raw video having predefined bandwidth requirements;
 obtaining, for the first PE device in the EVPN multihoming group, a first device address and a first network object identifier; 65

18

obtaining, for the second PE device in the EVPN multihoming group, a second device address and a second network object identifier;
 calculating a first normalized hash value by:
 computing a first hash value using the first device address and the first network object identifier; and
 normalizing the first hash value to a range between 0 and 1;
 calculating a second normalized hash value by:
 computing a second hash value using the second device address and the second network object identifier; and
 normalizing the second hash value to a range between 0 and 1;
 calculating a first score for the first PE device-by:
 taking the negative of a first capacity value of the first PE device; and
 dividing the negative of the first capacity value by a logarithm of the first normalized hash value;
 calculating a second score for the second PE device by:
 taking the negative of a second capacity value of the second PE device; and
 dividing the second negative of the capacity value by a logarithm of the second normalized hash value;
 if the first score is higher than the second score, designating the first PE device as the DF; and
 if the second score is higher than the first score, designating the second PE device as the DF.
 2. The method of claim 1, wherein the first network object identifier and the second network object identifier are Virtual Local Area Network (VLAN) identifiers.
 3. The method of claim 1, wherein the first network object identifier and the second network object identifier are Bridge Domain (BD) identifiers.
 4. The method of claim 1, wherein the first network identifier and the second network object identifier are EVPN Instance (EVI) identifiers.
 5. The method of claim 1, wherein the first network identifier and the second network object identifier are Ethernet Segment(ES) identifiers.
 6. The method of claim 5, wherein the first and second device addresses are Internet Protocol (IP) addresses.
 7. The method of claim 1, wherein if the first score is equal to the second score, evaluating the numeric value of the first device address and the second device address, and designating the PE device associated with the lower numeric value as the DF.
 8. The method of claim 1, wherein the first capacity value is a first link bandwidth and the second capacity value is a second link bandwidth.
 9. The method of claim 1, further comprising:
 at the second PE device:
 calculating the first normalized hash value by:
 computing the first hash value using the first device address and the first network object identifier of the first PE device; and
 normalizing the first hash value to a range between 0 and 1;
 calculating the second normalized hash value by:
 computing the second hash value using the second device address and the second network object identifier of the second PE device; and
 normalizing the second hash value to a range between 0 and 1;
 calculating a first score for the first PE device by:
 taking the negative of a first capacity value of the first PE device; and

19

dividing the negative of the first capacity value by a logarithm of the first normalized hash value;
 calculating a second score for the second PE device by:
 taking the negative of a second capacity value of the second PE device; and
 dividing the second negative of the capacity value by a logarithm of the second normalized hash value;
 if the first score is higher than the second score, designating the first PE device as the DF; and
 if the second score is higher than the first score, designating the second PE device as the DF.

10. A first provider edge (PE) device, comprising:
 one or more processors and one or more network connections; and
 wherein the first PE device is configured to perform operations comprising:
 receiving a request to route incoming traffic through one or more links within the EVPN;
 determining that the incoming traffic comprises raw video having predefined bandwidth requirements;
 calculating a first hash value using a first device address and a first network object identifier;
 calculating a second hash value using a second device address and a second network object identifier;
 normalizing the first hash value to a range between 0 and 1;
 normalizing the second hash value to a range between 0 and 1;
 assigning a first weight to the first PE device;
 assigning a second weight to the second PE device;
 generating a first score using the equation $-w1/\log(\text{hashn}(\text{PE1}))$, wherein $w1$ is the first weight, $\text{hashn}(\text{PE1})$ is the first normalized hash value, and $\log()$ denotes taking a logarithm of the first normalized hash value, and assigning the first score to the first PE device;
 generating a second score using the equation $-w2/\log(\text{hashn}(\text{PE2}))$, where $w2$ is the second weight, $\text{hashn}(\text{PE2})$ is the second normalized hash value, and $\log()$ denotes taking a logarithm of the second normalized hash value, and assigning the second score to the second PE device;
 if the first score is higher than the second score, designating the first PE device as a designated forwarder (DF); and
 if the second score is higher than the first score, designating the second PE device as the DF.

11. The first PE device of claim 10, wherein the first and second network object identifiers include Virtual Local Area Network (VLAN) identifiers.

12. The first PE device of claim 10, wherein the first and second network object identifiers include Bridge Domain (BD) identifiers.

13. The first PE device of claim 10, wherein the first and second network object identifiers include EVPN instance (EVI) identifiers.

14. The first PE device of claim 10, wherein the first and second network object identifiers include Ethernet Segment (ES) identifiers.

15. The first PE device of claim 10, wherein the operations performed at the first PE device further comprise:
 if the first score is equal to the second score, evaluating the numeric value of the first device address and the second device address; and
 assigning the PE device associated with the lower numeric device address value as the DF.

20

16. The first PE device of claim 15, wherein the first and second device addresses are IP addresses.

17. The first PE device of claim 10, wherein the first weight value is a first link bandwidth capacity and the second weight value is a second link bandwidth capacity.

18. The first PE device of claim 10, wherein the first PE device is configured to perform additional operations comprising:
 assigning an updated weight to the first PE device;
 calculating an updated score for the first PE device using the equation $-wU/\log(\text{hashn}(\text{PE1}))$, where wU is the updated weight, $\text{hashn}(\text{PE1})$ is the first normalized hash value, and $\log()$ denotes taking the logarithm of the value;
 assigning the updated score to the first PE device;
 if the updated score is higher than the second score, designating the first PE device as a designated forwarder (DF); and
 if the second score is higher than the updated score, designating the second PE device as the DF.

19. A system comprising the first PE device of claim 10, the system further comprising a second PE device in the EVPN multicast group with the first PE device, wherein the second PE device comprises:
 one or more processors and one or more network connections; and
 wherein the second PE device is configured to perform operations comprising:
 calculating the first hash value using the first device address and the first network object identifier;
 calculating the second hash value using the second device address and the second network object identifier;
 normalizing the first hash value to a range between 0 and 1;
 normalizing the second hash value to a range between 0 and 1;
 assigning the first weight to the first PE device;
 assigning the second weight to the second PE device;
 generating the first score using the equation $-w1/\log(\text{hashn}(\text{PE1}))$, where $w1$ is the first weight, $\text{hashn}(\text{PE1})$ is the first normalized hash value, and $\log()$ denotes taking the logarithm of the value, and assigning the first score to the first PE device;
 generating the second score using the equation $-w2/\log(\text{hashn}(\text{PE2}))$, where $w2$ is the second weight, $\text{hashn}(\text{PE2})$ is the second normalized hash value, and $\log()$ denotes taking the logarithm of the value, and assigning the second score to the second PE device;
 if the first score is higher than the second score, designating the first PE device as the DF; and
 if the second score is higher than the first score, designating the second PE device as the DF.

20. One or more computer-readable non-transitory storage media containing instructions that, when executed by one or more processors at a first provider edge (PE) device of a plurality of PE devices in the same EVPN multicast group ($\text{PE1} \dots \text{PEn}$), cause the first provider edge device (PE1) to perform operations comprising:
 receiving a request to route incoming traffic through one or more links within the EVPN;
 determining that the incoming traffic comprises raw video having predefined bandwidth requirements;
 deriving and assigning a device score for each PE device i (PEi) from 1 to n of the plurality of PE devices, wherein deriving and assigning a device score for PEn comprises:

21

calculating a PE device hash value for PE_i using a PE device address for PE_i and at least one network object identifier;
 calculating a normalized PE device hash value for PE_i by normalizing the PE device hash value for PE_i to a range between 0 and 1;
 assigning a PE device weight w_i to PE_i;
 generating a PE device score for PE_i (score_i) using the equation $\text{score}_i = -w_i / \log(\text{hashn}(\text{PE}_i))$, where w_i is the PE device weight assigned to the PE device i , $\text{hashn}(\text{PE}_i)$ is the normalized PE device hash value for PE_i, and $\log(\)$ denotes taking a logarithm of the normalized PE device hash value for PE_i; and
 assigning the PE device score score_i to PE_i of the plurality of PE devices;
 selecting the PE device of the plurality of PE devices associated with the highest assigned PE device score; and
 designating the selected PE device as a designated forwarder (DF).
 21. The computer-readable storage media of claim 20, wherein the at least one network object identifier includes a Virtual Local Area Network (VLAN) identifier.
 22. The computer-readable storage media of claim 20, wherein the at least one network object identifier includes a Bridge Domain (BD) identifier.
 23. The computer-readable storage media of claim 20, wherein the at least one network object identifier includes an EVPN instance (EVI) identifier.
 24. The computer-readable storage media of claim 20, wherein the operations performed at PE1 further comprise:
 after having derived and assigned the device score for each PE_i, if there is no highest assigned PE device score because the two highest PE device scores are equal:
 evaluating the numeric value of the PE device addresses associated with the PE devices having the two highest PE device scores; and

22

designating the PE device with the device address with the lower numeric value as the DF.
 25. The computer-readable storage media of claim 24, wherein the PE device addresses are Internet Protocol (IP) addresses.
 26. The computer-readable storage media of claim 20, wherein the PE device weight assigned to each PE device is a link bandwidth capacity associated with the PE device.
 27. The computer-readable storage media of claim 20, wherein the operations performed at PE1 further comprise:
 assigning an updated weight to PE1;
 calculating an updated score for PE1 using the equation $-wU / \log(\text{hashn}(\text{PE}_1))$, where wU is the updated weight, $\text{hashn}(\text{PE}_1)$ is the normalized hash value for PE1, and $\log(\)$ denotes taking a logarithm of the normalized hash value for PE1;
 assigning the updated score to PE1; and
 if the updated score for PE1 is the highest PE device score, designating PE1 as the DF.
 28. The computer-readable storage media of claim 20, wherein:
 at least one of the computer-readable storage media is associated with PE1 and at least one of the computer-readable storage media is associated with a second PE device (PE2) of the plurality of PE devices; and
 PE1 and PE2 each independently perform the same operations.
 29. The computer-readable storage media of claim 20, wherein calculating a PE device hash value for PE_i using a PE device address for PE_i and at least one network object identifier uses the equation $\text{hash}(\text{O}_i * \text{D}_i)$, where O_i is the at least one network object identifier and D_i is the PE device address for PE_i, and $*$ denotes a function combining the values for O_i and D_i .
 30. The computer-readable storage media of claim 29, where the function combining the values for O_i and D_i is a concatenation function.

* * * * *